Problem Set 4C

Scalar product of vectors

Do these after the slides for Section 4C. Textbook Exercise 4C: Questions 5, 9, 10, 11. Answers are on the last page. Show full working in your exercise book.

Skills

1 Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ and $\mathbf{c} = -3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. Find:

(a) $\mathbf{a} \cdot \mathbf{b}$

(b) $\mathbf{b} \cdot \mathbf{c}$

(c) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$

(d) $(\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}$

2 Find, correct to one decimal place, the angle between $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $3\mathbf{i} + 4\mathbf{k}$.

3 Find t if $2\mathbf{i} + t\mathbf{j} - \mathbf{k}$ is perpendicular to $t\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$.

4 Find the values of t for which $t\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ is perpendicular to $t\mathbf{i} + t\mathbf{j} + 3\mathbf{k}$.

5 Given $|\mathbf{a}| = 4$, $|\mathbf{b}| = 3$ and $\mathbf{a} \cdot \mathbf{b} = 6$, find:

(a) the angle between $\mathbf{a}$ and $\mathbf{b}$

(b) $|\mathbf{a} + \mathbf{b}|$

(c) $(\mathbf{a} - 2\mathbf{b}) \cdot (\mathbf{a} + \mathbf{b})$

Exam style

6 For the points $A(2, 0, 1)$, $B(3, 2, -1)$ and $C(0, 1, 3)$, find $\cos(\angle ABC)$ and the exact area of triangle ABC.

7 Show that if $|\mathbf{a}| = |\mathbf{b}|$, then $\mathbf{a} + \mathbf{b}$ is perpendicular to $\mathbf{a} - \mathbf{b}$.

8 Let $\mathbf{a} = 3\mathbf{i} + 4\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$. Find the value of λ for which $\mathbf{a} + \lambda\mathbf{b}$ is perpendicular to $\mathbf{b}$.

Problem Set 4C: answers

1 (a) -8 (b) 3 (c) -13 (d) -8

2 $\cos\theta = -\frac{1}{3}$, $\theta = 109.5°$

3 $t = \frac{4}{5}$

4 $t = 2$ or $t = -3$

5 (a) $60°$ (b) $\sqrt{37}$ (c) -8

6 $\cos(\angle ABC) = \dfrac{13}{3\sqrt{26}} = \dfrac{\sqrt{26}}{6}$; area $= \dfrac{\sqrt{65}}{2}$

7 $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2 = 0$

8 $\mathbf{a}\cdot\mathbf{b} + \lambda|\mathbf{b}|^2 = 10 + 9\lambda = 0$, so $\lambda = -\frac{10}{9}$